Daily Review



1. Air from either valve gets through: which control?
2. Two valves in a line: which control?
3. Fault-finding: check 1, every time?
4. Double-acting: what brings the rod back?
5. A system diagram: input, process and…

OR

AND

Regulator at 0

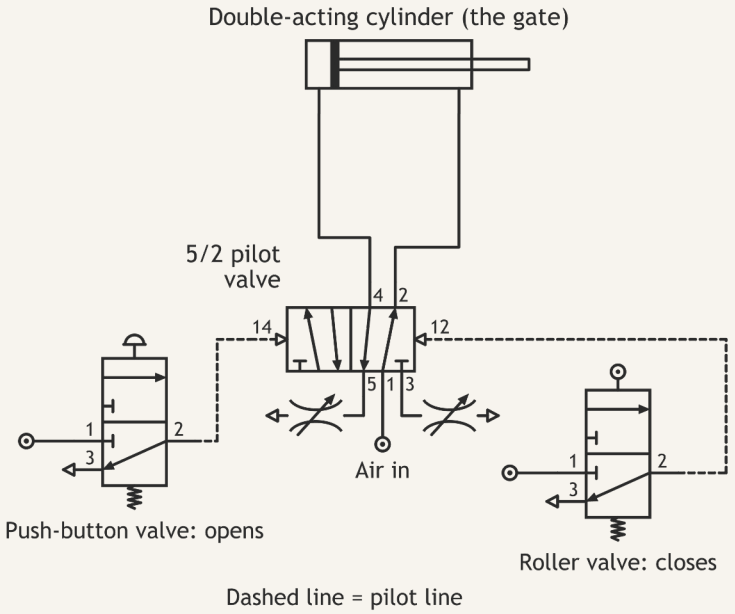
Air

Output

Level Up

The Big Gate

A small push, a big valve, a heavy gate.



S3 Engineering Science

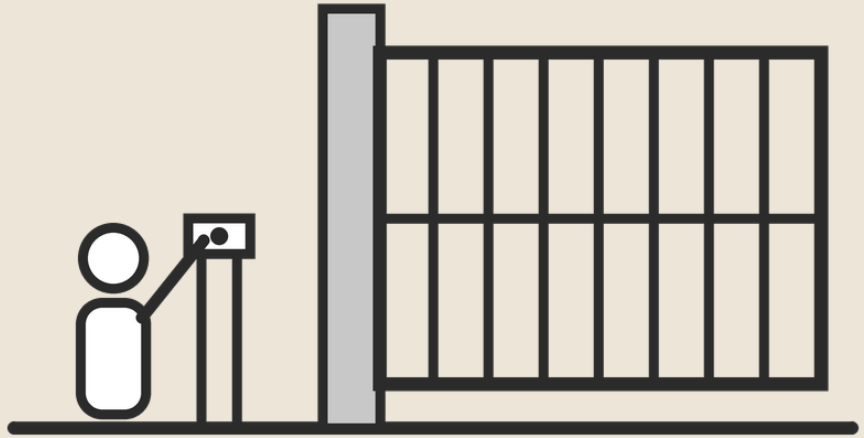
The Job: The Big Gate

i When the button is pressed, the gate opens.

ii When the button is let go, the gate stays open.

iii When the roller is pressed, the gate closes.

Predict: button and roller at once. The gate?



By the end of this lesson…



1. I can build and test the big gate from a bench card
2. I can say what a pilot valve remembers
3. I can read a pilot line and a pilot actuator
4. I can name the motion when a rod goes out and back

Endorsement

Sheet E1 (resit: E2)

3 questions, on your own.
You need 3 out of 3.
Then the kit check with the teacher.

SAFETY No endorsement, no air in Lessons 9 to 11.

2 min

Not Endorsed Yet? Your Route

Tasks 1, 3

build and test it in the sim

Pilot symbol

everyone learns it

Tasks 5-6

the same questions

Resit

sheet E2



THE BUILD

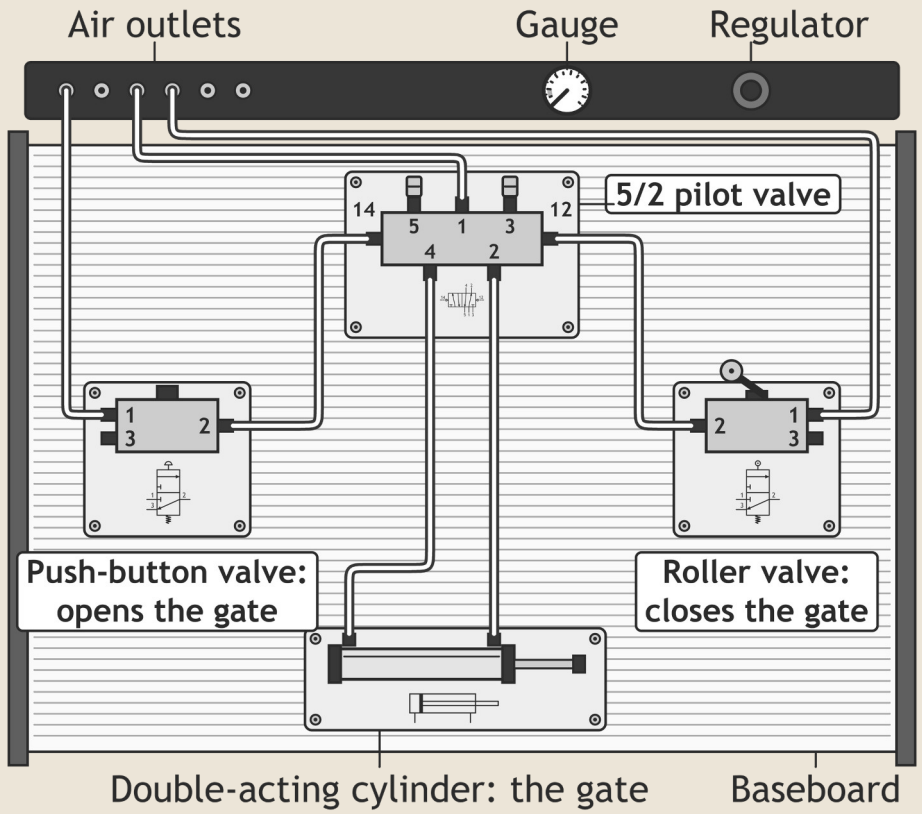
Build the Big Gate

Endorsed? Simulate, Air Permit, then the kit.

What to Collect

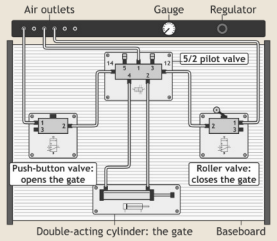
Builder and Safety: one set per team.

1 Bench card 7, tubing
2 Manifold (air supply)
3 5/2 pilot valve
4 Button valve, roller valve
5 Double-acting cylinder
6 2 exhaust restrictors
7 Goggles, one each



Simulate and Build

Tasks 1-2 (B6A-L7-p23)

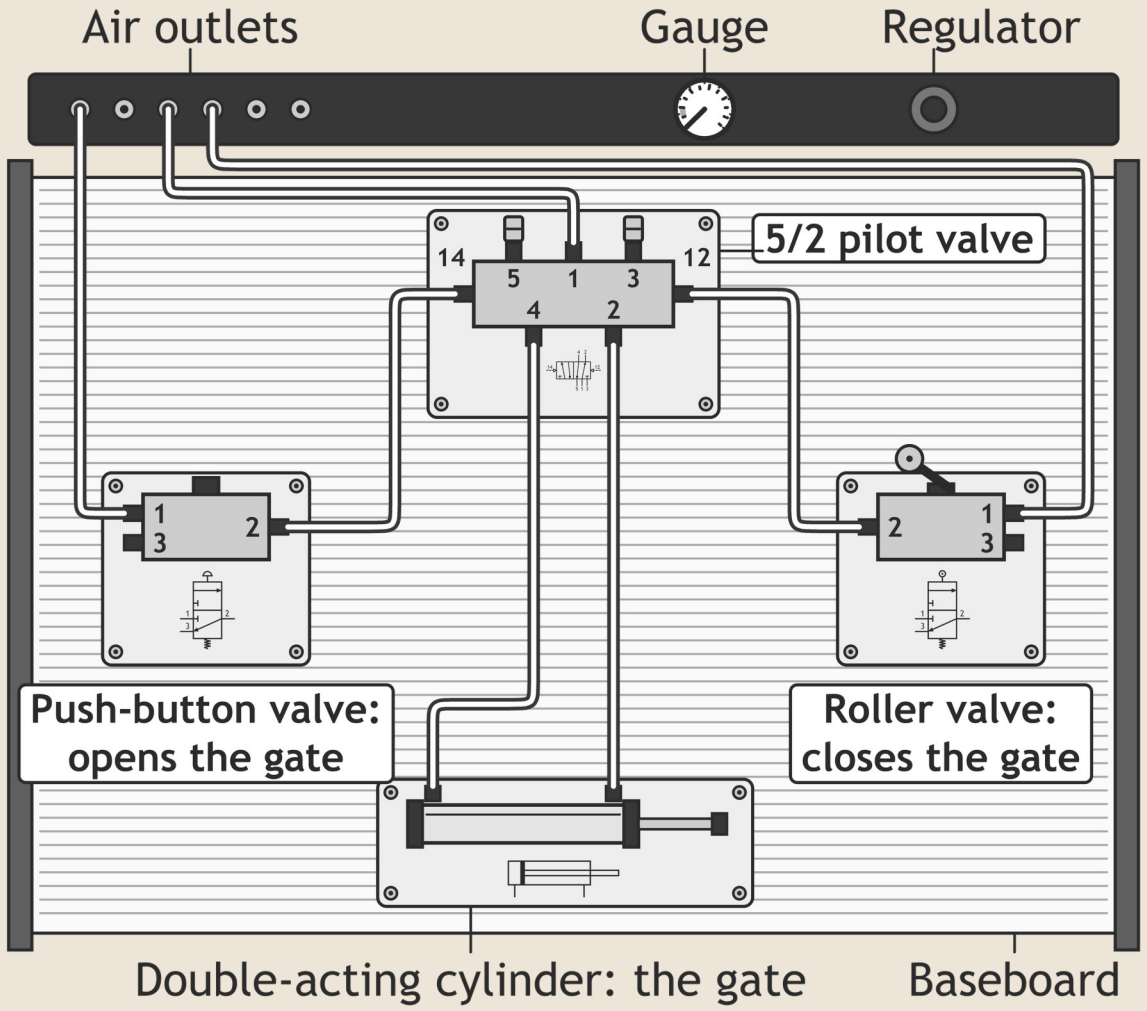


Tester and Recorder: start on the computer.
Builder and Safety: parts and bench card 7.
Safety, then teacher: check the Air Permit.

SAFETY Bolt the roller down first, with the air off.

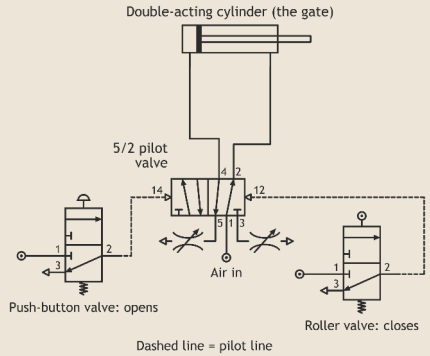
18 min

Build It: Bench Card 7



Find Out: Does It Remember?

Tasks 3-4 (B6A-L7-p23)



Task 3: try every row. Write what the rod does.

Task 4: compare the sim and the kit.

SAFETY Hands clear: the rod can move by itself.

6 min

Test It: What Did You Find? (B6A-L7-p23)

Job	What you do	What the rod does
i, ii	Press the button, then let go.	Out, and stays out
iii	Press the roller, then let go.	In, and stays in
-	Press both at once.	It does not move
How?	What moved the big valve?	Air from the button

A Small Push, a Big Valve

Button valve

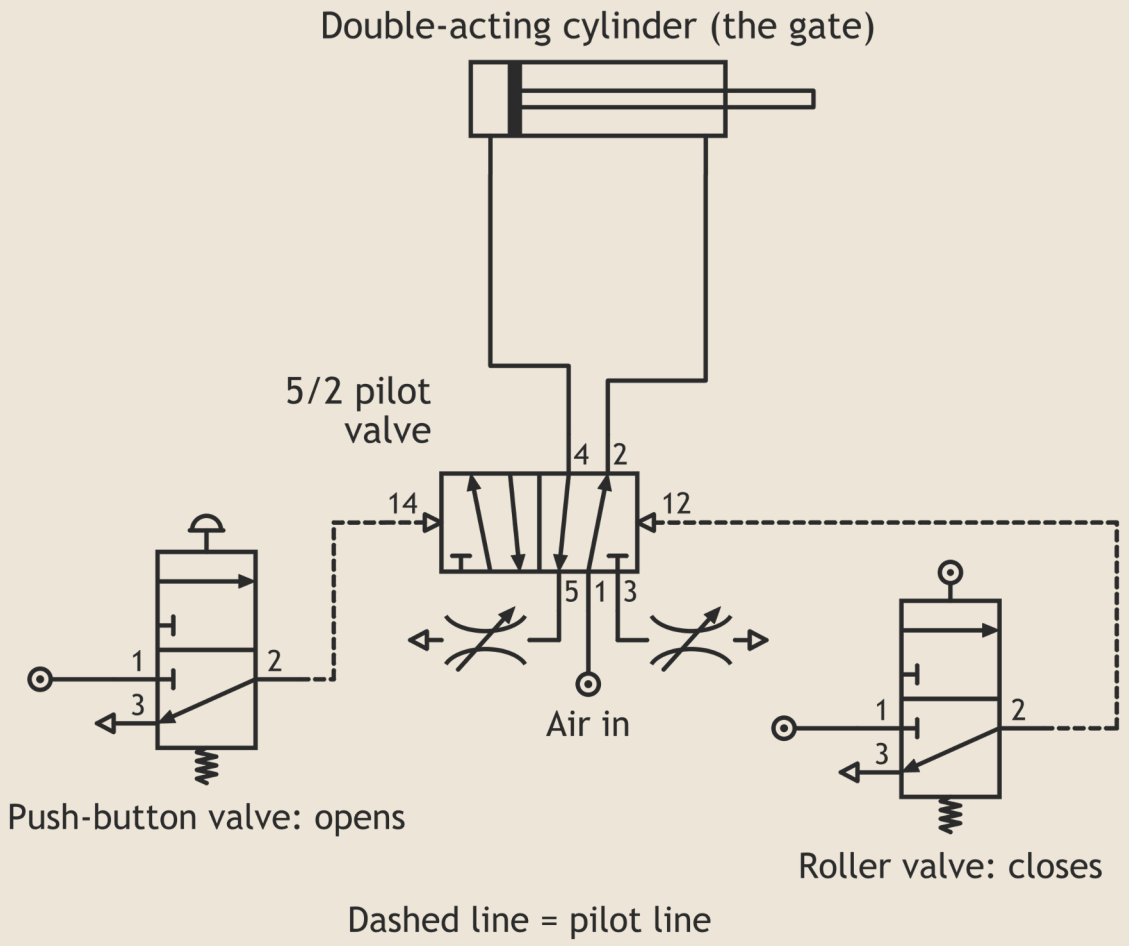
sends a little pilot air

Pilot air

pushes the big valve

Big valve

sends the main air



The Pilot Symbol

Core

Pilot line

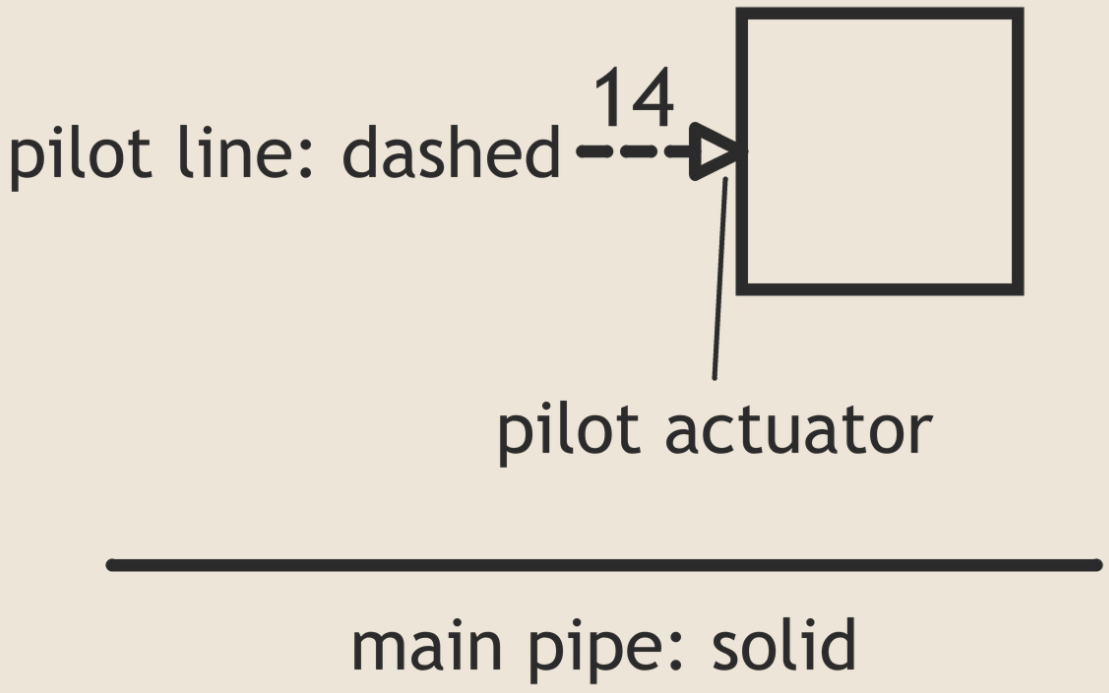
dashed: signal air

Pilot actuator

a triangle at the valve end

Main pipe

solid



The 5/2 Pilot Valve

14

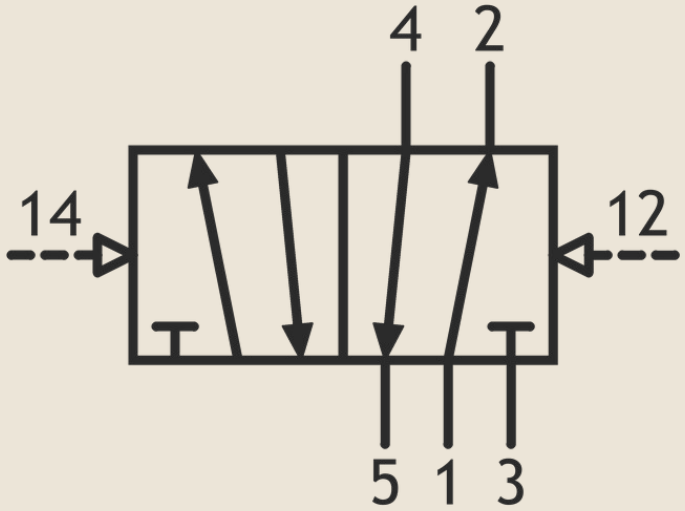
air here: rod goes out

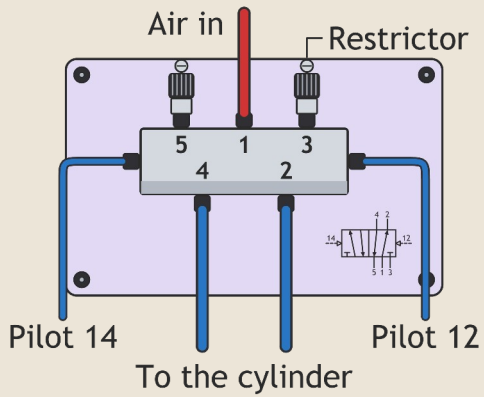
12

air here: rod comes in

No pilot air

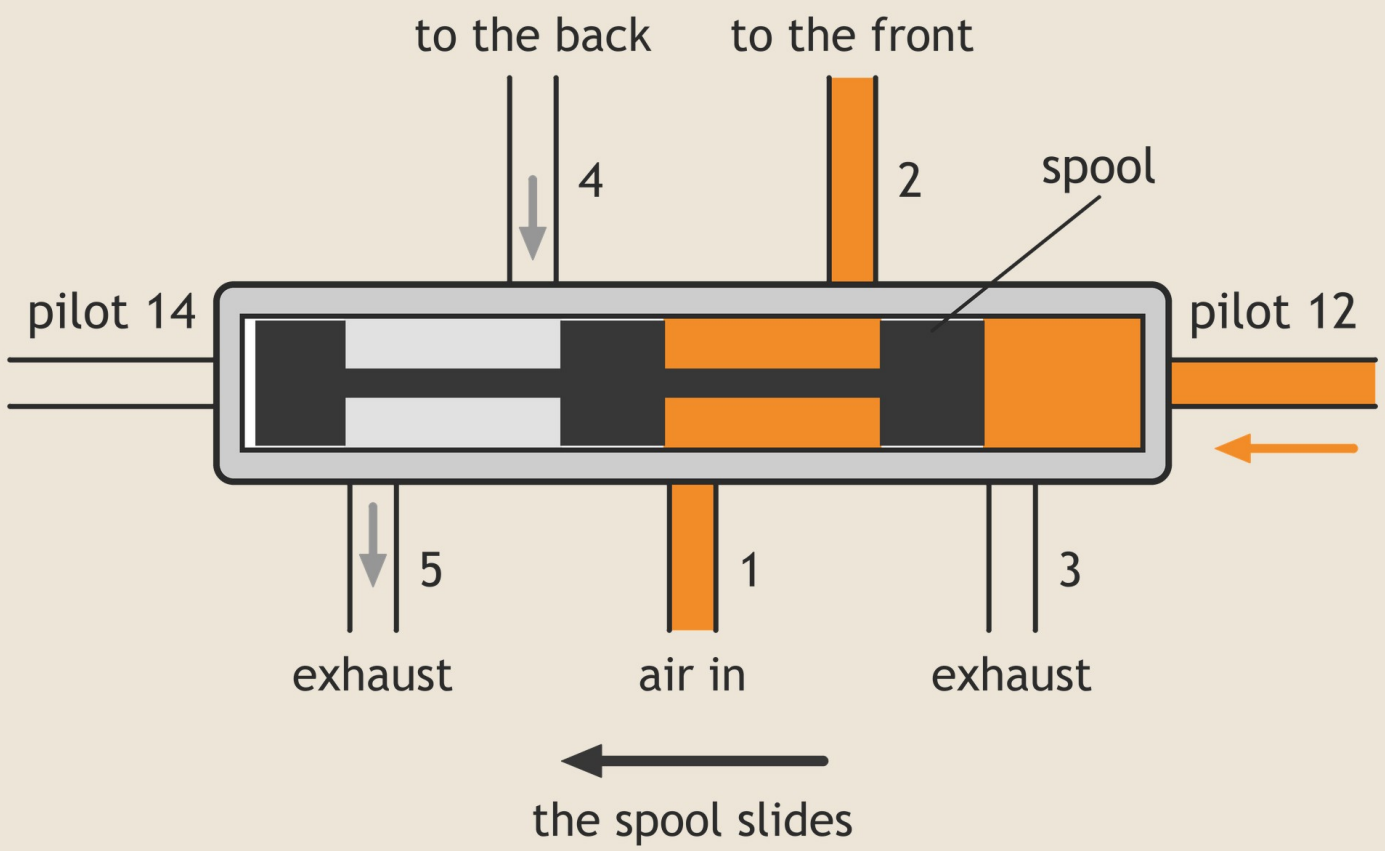
it stays put





The real valve: one per group.

Inside: It Remembers



1 Air to 14 slides the spool. Air 1 to 4: rod out.

2 Let go: no pilot air. The spool stays: rod stays out.

3 Air to 12 slides it back. Air 1 to 2: rod in.

Out and Back: Reciprocating

State the type of motion a pneumatic piston produces when it outstrokes and then instrokes.

Hint

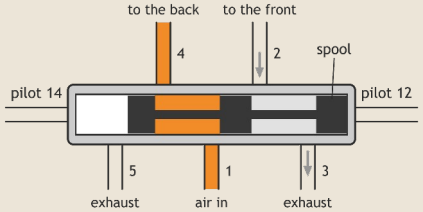
button: out; roller: back

Reciprocating (1)

Motion name	Graphic
Linear	
Reciprocating	

Questions and Key Words

Tasks 5-6 (B6A-L7-p24)



Task 5: Questions 1 to 3 on your own.

Task 6: Key words.

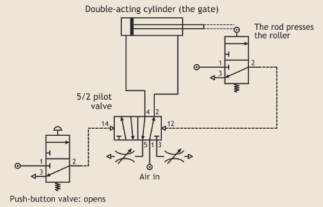
6 min

Answers: Tasks 5-6 (B6A-L7-p24)

Question	Mark 1	Mark 2
Q1 (2 marks)	Only sends pilot air	Pilot air moves valve
Q2 (1 mark)	The rod stays out	-
Q3 (1 mark)	Regulator at 0	-
Key words 1-2	pilot	14
Key words 3-4	back	remembers

Optional: A Gate That Closes Itself

Task 8, card 7 side 2



Regulator to 0. Roller to the rod end.
Press the button once. Watch.
Set the restrictors: out fast, back slowly.

SAFETY Move the roller only with the regulator at 0.

8 min

Answers: Task 8 (B6A-L7-p25)

1 Q4 Press the button once. The gate…

Opens, then closes by itself

Plenary: The Big Gate

1	A pilot line is drawn…	Dashed
2	Air to pilot port 14: the rod…	Goes out
3	Let go of the button. The rod…	Stays out
4	Why can a small button work a big valve?	It only sends pilot air
5	Out, then back: the motion is…	Reciprocating

Look back — by the end of this lesson…



1. I can build and test the big gate from a bench card
2. I can say what a pilot valve remembers
3. I can read a pilot line and a pilot actuator
4. I can name the motion when a rod goes out and back

LOOKING AHEAD

Lesson 10: The Pirate Hides by Itself

A pirate pops up, waits, then hides by itself. How?

Credits

The job and the kit: Sources: job card and kit drawings by R Stewart.

Big gate circuit: The big gate circuits adapted from the Matrix CP2080 Pneumatics Essentials Worksheet 5 (© Matrix Technology Solutions Limited). Drawings by R Stewart.

Real part drawings: Sources: kit drawings by R Stewart, from the Matrix CP2080 kit.

Out and Back: 2024 Q15(d), 2021 Q1 graphics: © Qualifications Scotland (SQA). N5 Engineering Science 2024, Question 15(d) and its marking instructions; the motion graphics from the 2021 marking instructions, Question 1.

R Stewart - October 2026